Exp No: 4
Date: 3.8.26

Setting up a local area Network

Aim: To design, setup & configure a LAN connecting a components through a network, verify design using the ping command and demonstrate resource/data sharing across the network.

Implementation in Cisco Packet Tracer:

Since physical hardware was limited, the LAN was also implemented and verified using Cisco Packet Tracer simulation software following the steps below.

Step 1: Place the Device

Open Cisco Packet Tracer Select 4 PC's onto workspace select & drag & One switch.

Step 2: Cable the Device

Click connection → Copper straight wire and connect all 4 PCs to switch via fast ethernet port.

Step 3: Assign IP address to PC's

Step 4: Click the switch → CLI tab, Enter the following Commands.

```
enable
configure terminal
interface vlan 1
ip address 10.1.1.5 255.0.0.0
no shutdown
exit
```

Step 5: Verify connectivity using ping

- Click a PC → Desktop tab → Command prompt
- Use the ping command to test connectivity to the other PC's e.g., ping 10.1.1.2.
- Successfully "Reply from" messages confirm connectivity

Step 6: Demonstrate Data Sharing (Server Simulation)

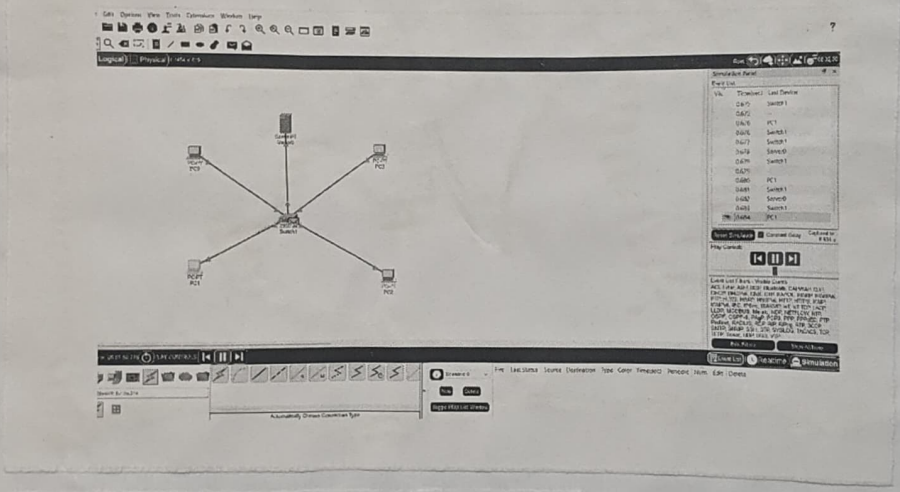
- Add a server device (End Device → Server-PT) and connect it to the Switch.
- Assign it a IP address via its Desktop → IP configuration tab.

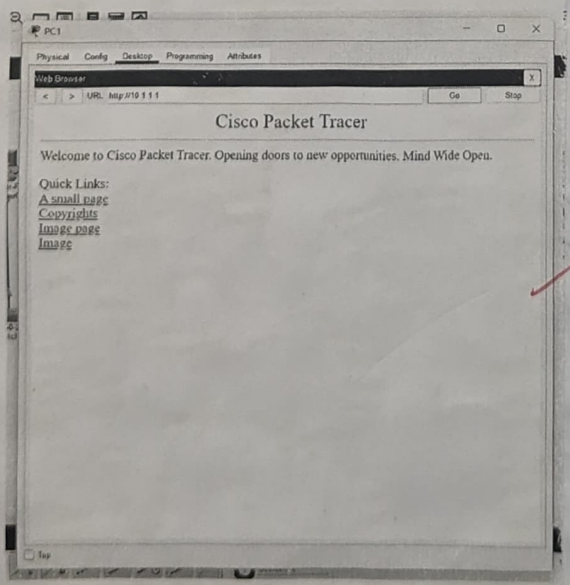
• Open the server's services (Config) tab & enable the HTTP service.

• From another PC, open the Web Browser (Desktop tab) and enter the server's IP address to load the page, confirming successful data transfer across the LAN.

Step 7: Observe Packet flow in Simulation Mode

Switch to Simulation mode, send a ping or web request between two devices and use the Play/Capture controls to step through the packet as it travels chop-by-chop through the switch, observing MAC-address-based forwarding.





Result: All PC's were connected and the Ping test between all devices returned successful replies, confirming end to end connectivity, the LAN is also functioning correctly for resource sharing.